



Risk-Adaptive Neurosymbolic Intent Planning for Optical Networks

A Pre-Deployment Decision Mechanism with Joint Semantic and QoT Assessment

Candidate: Felipe Abadía

Academic Advisor: Prof. Massimo Tornatore

Dipartimento di Elettronica, Informazione e Bioingegneria — Politecnico di Milano

OUTLINE

[Problem 1] The Optical Intent Planning Bottleneck

Physical-layer constraints, token saturation, and hallucinated routing in optical backbones

[Architecture] Neurosymbolic Intent Planning Pipeline

Decoupling probabilistic reasoning (NL to PDDL) from deterministic solvers and physics tools

[Mechanism] Pre-Deployment Risk Gates: Semantic & Physical Validation

Sequential fail-fast decision via Layer 1/2 semantic gate (U_sem) and GN-model QoT gate

[Evaluation] Experimental Testbed Validation on 17-Node Optical Topology

Benchmarking safety, human intervention reduction, and orchestration latency against baselines

[Outlook] Key Takeaways, System Guarantees & Future Directions

Summary of thesis contributions, operational guarantees, and extension to joint compute scheduling

Motivation: The Vision of Intent-Based Optical Networks

Operational Paradigm Shift

- Optical backbones carry terabits of core traffic across ROADM networks
- Traditional workflow: manual CLI scripts and complex RESTConf payloads
- Human configuration delays lightpath provisioning by hours or days
- Goal: Transition to autonomous Intent-Based Networking (IBN)
- Allow network operators to express high-level operational goals naturally

The Operational Promise

- High-level abstraction: specify WHAT is needed, not HOW to configure it
- Example: 'Establish a 400G lightpath between Milan and Rome avoiding L2'
- Autonomous translation into validated physical lightpaths
- Reduces human configuration error across multi-vendor optical links
- Critical challenge: Optical networks do not tolerate probabilistic errors

Illustrative Failure: Why Standard LLMs Break Optical Backbones

Challenge 1: Token Budget Saturation

- Full optical topology payloads (RESTConf JSON) exceed LLM context budgets
- In a 100-node core network, telemetry dumps consume tens of thousands of tokens
- Induces severe 'lost-in-the-middle' attention degradation
- Result: The LLM drops explicit user constraints such as link exclusion rules
- High API token cost and unpredictable prompt execution times

Challenge 2: Hallucinated Physics

- LLMs are probabilistic text predictors, not optical physics calculators
- Incapable of computing Generalized Signal-to-Noise Ratio (GSNR)
- Ignore nonlinear fiber Kerr effects and EDFA noise accumulation
- Result: Proposes lightpaths with unfeasible optical Quality of Transmission
- Causes severe traffic drop or optical controller rejection upon deployment

Problem Statement: Inputs, Constraints & Objectives

1. Given Inputs

- Unstructured Natural Language intent from operator
- Physical topology graph $G(V, E)$ via RESTConf
- Link fiber parameters: span lengths, attenuation
- EDFA amplifier gains and noise figures
- Transponder specs: baud rates, modulation formats

2. Constraints

- Semantic alignment: formal model matches intent
- Uncertainty bounded: $U_{\text{sem}} \leq \tau_{\text{sem}}$
- Optical GSNR exceeds modulation threshold
- Receiver power satisfies sensitivity: $P_{\text{rx}} \geq P_{\text{min}}$
- Zero spectral overlap and wavelength collision

3. Objectives

- Zero unfeasible routes reaching controller
- Fail-fast pre-deployment validation pipeline
- Selective, risk-proportional HITL engagement
- Sub-second deterministic computation time
- Auditable planning report for network engineers

Proposed Solution: Neurosymbolic Decoupling

Core Architectural Principle

- LLMs Reason, Deterministic Tools Calculate
- Prohibit LLMs from performing graph routing or arithmetic
- Constrain the LLM strictly to formal linguistic translation
- Natural Language is parsed into formal PDDL constraints
- Path computation delegated to Yen's KSP graph algorithms
- Physical validation delegated to analytical GN-model engine

Four Core Contributions

- 1. Neurosymbolic Pipeline: High-accuracy NL-to-PDDL translation
- 2. Scoped Optical GraphRAG: k-hop neighborhood extraction
- 3. Pre-Deployment Risk Gates: Joint U_{sem} and QoT validation
- 4. Production Orchestration: LangGraph StateGraph engine
- Guarantees provable physical safety before configuration push

End-to-End System Architecture & Pipeline Flow

Phase 1 Intent Ingest & Optical RAG

Enrich operator intent with ITU-T optical grid standards and transponder modes

Phase 2 PDDL Intent Parsing

Translate enriched intent into formal Planning Domain Definition Language constraints

Phase 3 Semantic Gate (U_{sem})

Evaluate CFG regex syntax and Reverse Prompting disagreement to catch hallucinations early

Phase 4 Symbolic Solver & GraphRAG

Extract scoped k-hop subtopology and generate K-Shortest Paths via NetworkX

Phase 5 QoT Physics Validation

Compute deterministic GSNR and receiver power P_{rx} using Python GN-model

Phase 6 Risk-Adaptive Decision Gate (RADG)

Execute piecewise decision function $D(U_{sem}, QoT_{valid}) \rightarrow \text{Auto-Approve} / \text{Replan} / \text{Clarify}$

Phase 7 Plan Synthesizer

Generate auditable planning report with verification traces and testbed deployment commands

Overcoming Token Saturation: Scoped Optical GraphRAG

The Problem: Topology Bloat

- Full topology dumps overwhelm LLM context windows
- Raw JSON contains excessive telemetry: ROADMs ports, EDFAs, fibers
- Causes severe attention degradation on key operational constraints
- Linear increase in inference latency and financial token cost
- Fragile prompt structures prone to truncation and context loss

Deterministic k-hop Subtopology Scoping

- Graph engine extracts only the k-hop neighborhood between endpoints
- Filters out irrelevant core subnets, links, and unused transponders
- Reduces prompt token payload by over 75%
- $O(V + E)$ graph traversal executes in sub-millisecond Python code
- Guarantees sharp LLM attention focus on active optical constraints

Overcoming Semantic Drift: Reverse Prompting & HITL

Two-Layer Semantic Uncertainty (U_{sem})

- Layer 1 (Structural): Context-Free Grammar (CFG) regex validator
- Instantly rejects hallucinated predicates or malformed PDDL syntax
- Layer 2 (Semantic): Reverse Prompting reconstructs NL from PDDL
- Independent LLM judge measures semantic divergence d_{sem} in $[0, 1]$
- Formal metric: $U_{\text{sem}} = 1$ if syntax invalid, else $U_{\text{sem}} = d_{\text{sem}}$

Fail-Fast HITL Clarification Loop

- Evaluated BEFORE invoking graph solvers or physical tools
- If $U_{\text{sem}} > \tau_{\text{sem}}$: Pipeline pauses via `LangGraph interrupt()`
- Prompts operator to clarify ambiguous parameters or missing nodes
- Eliminates infinite trial-and-error conversational loops
- Guarantees that downstream tools receive mathematically verified intent

Pre-Deployment Risk Gate: The RADG Decision Function

1. State Space & Inputs

- Sequential pre-deployment risk evaluation
- Signal 1: Semantic Uncertainty U_{sem} in $[0, 1]$
- Signal 2: Binary QoT feasibility QoT_{valid}
- Action space $A = \{\text{approve, clarify, replan}\}$
- Prevents deployment of unverified network plans

2. Piecewise Formulation

- $D = \text{clarify}$ IF $U_{\text{sem}} > \tau_{\text{sem}}$
- $D = \text{replan}$ IF $U_{\text{sem}} \leq \tau_{\text{sem}}$ AND $QoT_{\text{valid}} = 0$
- $D = \text{approve}$ IF $U_{\text{sem}} \leq \tau_{\text{sem}}$ AND $QoT_{\text{valid}} = 1$
- Enforces fail-fast sequential decision tree
- Physics never evaluated if semantics fail

3. Action Guarantees

- Auto-Approve: Autonomous zero-touch deployment
- Suggest Replan: Prompts operator to relax constraints
- Clarify: Resolves missing intent parameters early
- Zero unverified configurations reach controller
- Drastically cuts human operator fatigue

Deterministic Physical Layer: GN-Model QoT Validation

Gaussian Noise (GN) Physical Engine

- Pure Python implementation: zero LLM arithmetic involvement
- Accumulated Amplified Spontaneous Emission (ASE) noise:
 - $P_{ASE} = (G - 1) * h * \nu * F * B_{ref}$
- Nonlinear Interference (NLI) noise power per span:
 - $P_{NLI} \sim \eta * P_{ch}^3$
- Models fiber dispersion, Kerr nonlinearity, and EDFA noise

Feasibility & Performance Criteria

- Generalized Signal-to-Noise Ratio (GSNR):
 - $GSNR = P_{ch} / (P_{ASE} + P_{NLI}) \geq GSNR_{threshold}$
- Receiver power budget check:
 - $P_{rx} = P_{launch} - A_{total} + G_{total} \geq P_{rx,min}$
- Execution speed: under 5 milliseconds per candidate route
- 100% deterministic reproducibility across physical spans

Experimental Setup & Testbed Environment

17-Node German Core Network

- Realistic telecom benchmark: 17 ROADMs and 26 fiber links
- Standard Single-Mode Fiber (SMF-28): $\alpha = 0.2 \text{ dB/km}$
- Dispersion parameter $D = 16.7 \text{ ps}/(\text{nm}\cdot\text{km})$, $\gamma = 1.2 / (W\cdot\text{km})$
- Amplified spans: dual-stage EDFAs with noise figure $F = 5.5 \text{ dB}$
- Dynamic link lengths ranging from 45 km to 350 km per span

Software Stack & Orchestration

- Orchestrator: LangGraph StateGraph with memory persistence
- Physics Engine: Deterministic Python GN-model validator
- Symbolic Solver: NetworkX Yen's KSP path generation
- Testbed Interface: RESTConf and Mock SDON testbed adapters
- Evaluated LLMs: Claude 3.5 Sonnet and GPT-4o via API

Evaluation Framework & Benchmark Scenarios

Comparative Baselines

- Baseline A (LLM-Only): Direct prompt-to-JSON generation
- Baseline B (Static Rule-Based): Strict regex + Always-on HITL
- Proposed: Neurosymbolic RADG with sequential gates

Test Scenarios (100 Demands)

- Nominal: Unambiguous requests with feasible routes
- Ambiguous: Missing constraints triggering U_sem
- Unfeasible: Long reaches violating GSNR thresholds
- Adversarial: Prompts designed to induce hallucinations

Evaluation Metrics

- Pre-deployment safety: Unfeasible route leakage %
- Operator fatigue: Human intervention reduction %
- Orchestration speed: End-to-end planning latency
- Solver efficiency: Graph search time vs topology scale

Key Findings & Pre-Deployment Guarantees

Pre-Deployment Safety Guarantee

- Zero physically unfeasible lightpaths reached the controller (100% safety)
- LLM-only baseline allowed up to 34% unfeasible routes to deploy
- Semantic gate caught 100% of malformed PDDL and syntax drift early
- Physical gate intercepted all GSNR and receiver power violations
- Eliminates trial-and-error reactive crash loops on live networks

Operational Efficiency & Latency

- Over 70% reduction in human intervention compared to always-on HITL
- Operator engaged only when genuine ambiguity exists ($U_{\text{sem}} > \tau_{\text{sem}}$)
- Deterministic solver and QoT evaluation execute in under 15 ms
- End-to-end orchestration latency averages under 2.5 s
- Proof that provable physical safety does not compromise speed

Conclusions & Main Takeaways

1. Decoupling is Essential

- LLMs must reason, tools must calculate
- Language models excel at PDDL semantic parsing
- Optical physics must remain 100% deterministic

2. Sequential Risk Gates

- Early semantic evaluation (U_{sem}) halts drift early
- Downstream GN-model ensures optical viability
- Guarantees 100% pre-deployment safety

3. Selective HITL Value

- Engaging humans proportionally to risk cuts fatigue by 70%
- Preserves operator trust through auditable reports
- Enables scalable autonomous optical operations

4. Production Ready

- LangGraph orchestrator tested on 17-node topology
- Sub-second deterministic compute time
- Extensible architecture for next-gen SDN

Future Outlook & Acknowledgments

Future Research Directions

- Joint Optical & Compute Scheduling: Co-scheduling optical paths with GPU jobs
- Multi-Band Optical Systems: Expanding GN-model to C+L band non-linearities
- Real-Time Telemetry Integration: Online closed-loop self-healing via telemetry
- Reinforcement Learning: Fine-tuning PDDL translations from operator feedback

Acknowledgments & Discussion

- Sincere thanks to Prof. Massimo Tornatore for his guidance and leadership
- Gratitude to the SDON Laboratory research group at Politecnico di Milano
- Thank you for your attention
- Open for questions and technical discussion